lab_10

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Load Data

```
library(skimr)
library(ggplot2)
library(dplyr)
```

```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```
library(ggrepel)
library(corrplot)
```

```
corrplot 0.95 loaded
```

```
url <- "https://raw.githubusercontent.com/fivethirtyeight/data/master/candy-power-ranking/can
candy.data <- read.csv(url, row.names = 1)
```

Summary

```r
n_type <- ncol(candy.data)
fruity <- sum(candy.data["fruity"])
```

> Q1. How many candy types are in the dataset?

There are ‘12 candy types.

> Q2. How many fruity candy types are in the datset?

38

```r
my_winp <- candy.data["Hershey's Kisses", "winpercent"]
kitkat_winp <- candy.data["Kit Kat", "winpercent"]
tootsie_winp <- candy.data["Tootsie Roll Snack Bars", "winpercent"]
```

> Q3. What is your favorite candy in the dataset and what is it’s winpercent value?

Hershey’s Kisses: 55.375454

> Q4. What is the winpercent value for “Kit Kat”?

76.7686

> Q5. What is the winpercent value for “Tootsie Roll Snack Bars”?

49.653503

```r
skim(candy.data)
```

Table 1: Data summary

Name	candy.data
Number of rows	85
Number of columns	12
Column type frequency:	
numeric	12
Group variables	None

Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
chocolate	0	1	0.44	0.50	0.00	0.00	0.00	1.00	1.00	
fruity	0	1	0.45	0.50	0.00	0.00	0.00	1.00	1.00	
caramel	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
peanutyalmondy	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
nougat	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
crispedricewafer	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
hard	0	1	0.18	0.38	0.00	0.00	0.00	0.00	1.00	
bar	0	1	0.25	0.43	0.00	0.00	0.00	0.00	1.00	
pluribus	0	1	0.52	0.50	0.00	0.00	1.00	1.00	1.00	
sugarpercent	0	1	0.48	0.28	0.01	0.22	0.47	0.73	0.99	
pricepercent	0	1	0.47	0.29	0.01	0.26	0.47	0.65	0.98	
winpercent	0	1	50.32	14.71	22.45	39.14	47.83	59.86	84.18	

> Q6. Is there any variable/column that looks to be on a different scale to the majority of the other columns in the dataset?

Yes, the winpercent column

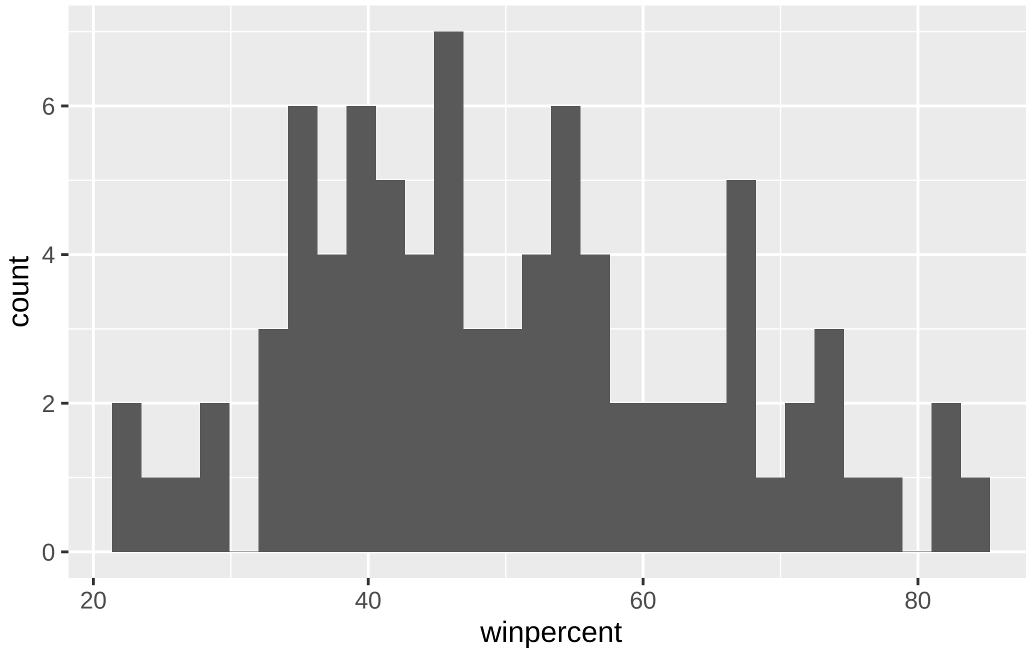
> Q7. What do you think a zero and one represent for the candy$chocolate column?

If a given candy is (1) or isn't (0) of type chocolate

> Q8. Plot a histogram of winpercent values

```
ggplot(candy.data) +
  aes(winpercent) +
  geom_histogram()
```

```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



Q9. Is the distribution of winpercent values symmetrical?

No, it's biased to the left

Q10. Is the center of the distribution above or below 50%?

Below 50%

```
chocolate <- candy.data |> filter(chocolate==1) |> select(winpercent)
ch_avg <- mean(chocolate$winpercent)

fruity <- candy.data |> filter(fruity==1) |> select(winpercent)
fr_avg <- mean(fruity$winpercent)

ans <- t.test(chocolate$winpercent, fruity$winpercent)
ans$p.value
```

```
[1] 2.871378e-08
```

```
candy.data |> arrange(-winpercent) |> head(5)
```

```
                          chocolate fruity caramel peanutyalmondy nougat
```

```
Reese's Peanut Butter cup        1       0       0             1       0
Reese's Miniatures               1       0       0             1       0
Twix                             1       0       1             0       0
Kit Kat                          1       0       0             0       0
Snickers                         1       0       1             1       1
                          crispedricewafer hard bar pluribus sugarpercent
Reese's Peanut Butter cup                0    0   0        0        0.720
Reese's Miniatures                       0    0   0        0        0.034
Twix                                     1    0   1        0        0.546
Kit Kat                                  1    0   1        0        0.313
Snickers                                 0    0   1        0        0.546
                          pricepercent winpercent
Reese's Peanut Butter cup        0.651   84.18029
Reese's Miniatures               0.279   81.86626
Twix                             0.906   81.64291
Kit Kat                          0.511   76.76860
Snickers                         0.651   76.67378
```

> Q11. On average is chocolate candy higher or lower ranked than fruit candy?

Chocolate candy is higher ranked 60.9215294 than fruity 44.1197414

> Q12. Is this difference statistically significant?

Yes the difference is statistically significant p-value = 2.8713778×10^{-8}

> Q13. What are the five least liked candy types in this set?

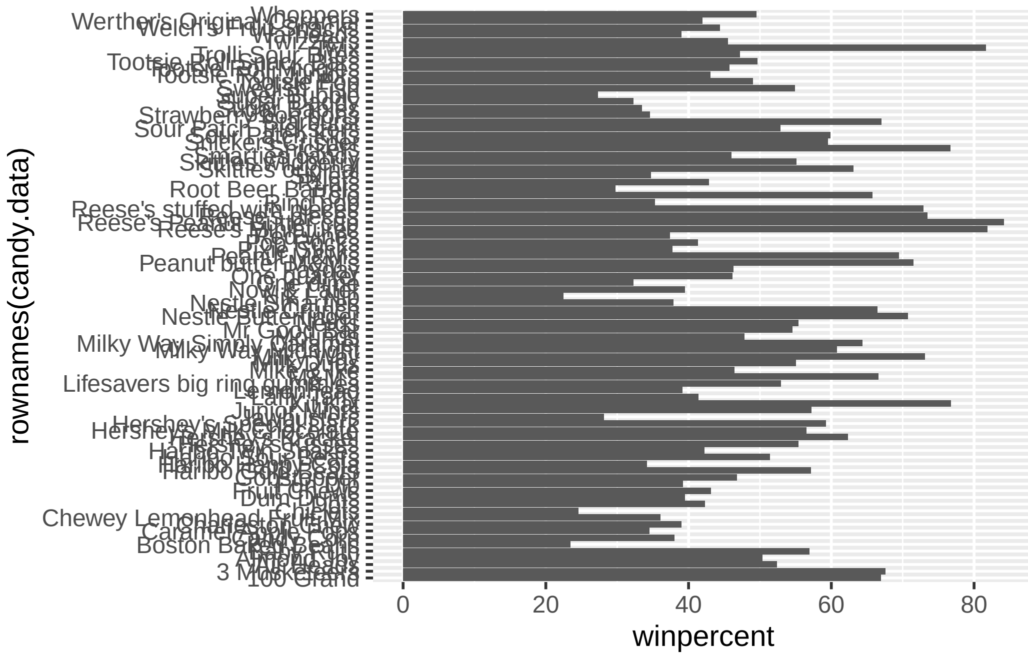
c(0, 0, 0, 0, 0), c(1, 0, 1, 1, 1), c(0, 0, 0, 0, 0), c(0, 1, 0, 0, 0), c(0, 0, 0, 0, 0), c(0, 0, 0, 0, 0), c(0, 0, 0, 0, 1), c(0, 0, 0, 0, 0), c(1, 1, 1, 0, 1), c(0.197, 0.31299999, 0.046, 0.162, 0.093000002), c(0.97600001, 0.51099998, 0.32499999, 0.116, 0.51099998), c(22.445341, 23.417824, 24.524988, 27.303865, 28.127439)

> Q14. What are the top 5 all time favorite candy types out of this set?

c(1, 1, 1, 1, 1), c(0, 0, 0, 0, 0), c(0, 0, 1, 0, 1), c(1, 1, 0, 0, 1), c(0, 0, 0, 0, 1), c(0, 0, 1, 1, 0), c(0, 0, 0, 0, 0), c(0, 0, 1, 1, 1), c(0, 0, 0, 0, 0), c(0.72000003, 0.034000002, 0.546, 0.31299999, 0.546), c(0.65100002, 0.27900001, 0.90600002, 0.51099998, 0.65100002), c(84.18029, 81.866257, 81.642914, 76.7686, 76.673782)

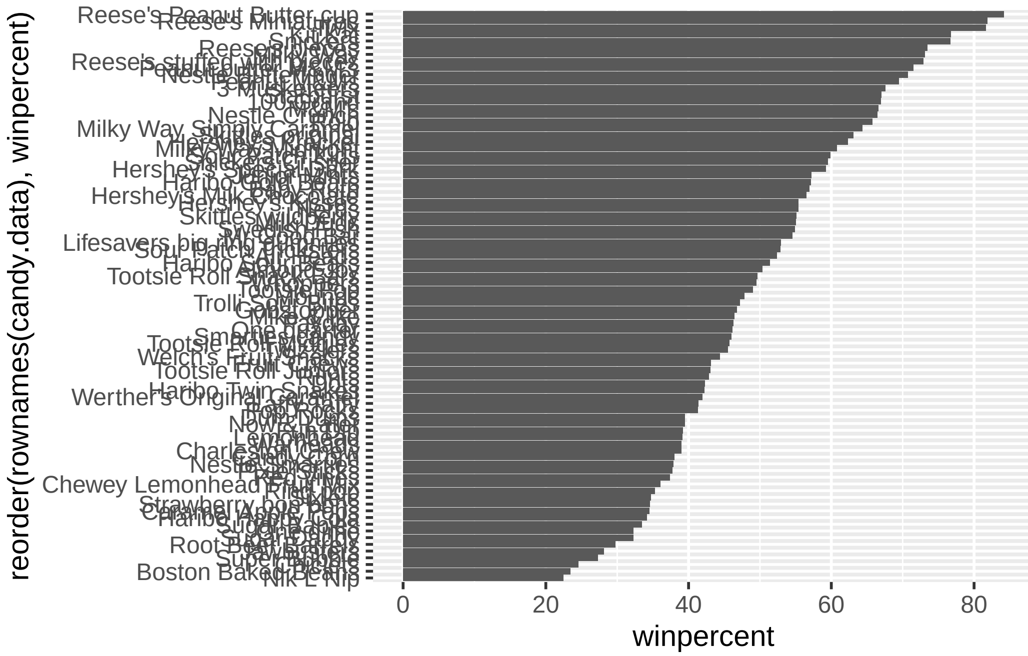
> Q15. Make a first barplot of candy ranking based on winpercent values.

```r
ggplot(candy.data) +
  aes(winpercent, rownames(candy.data)) +
  geom_col()
```



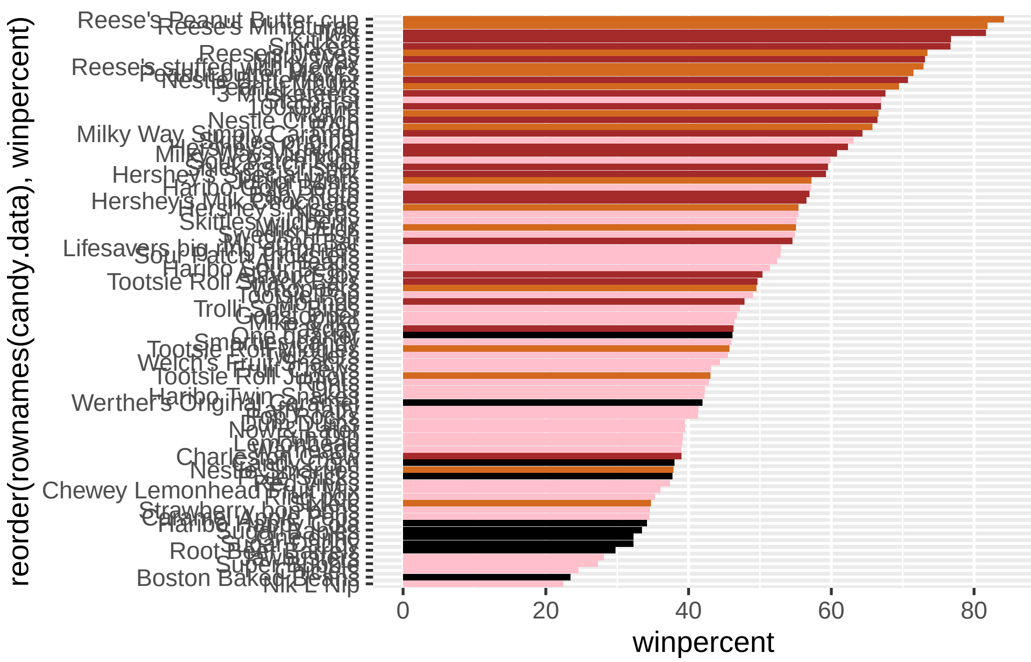
Q16. This is quite ugly, use the reorder() function to get the bars sorted by winpercent?

```
ggplot(candy.data) +
  aes(winpercent, reorder(rownames(candy.data), winpercent)) +
  geom_col()
```




```r
my_cols=rep("black", nrow(candy.data))
my_cols[as.logical(candy.data$chocolate)] = "chocolate"
my_cols[as.logical(candy.data$bar)] = "brown"
my_cols[as.logical(candy.data$fruity)] = "pink"

ggplot(candy.data) +
  aes(winpercent, reorder(rownames(candy.data),winpercent)) +
  geom_col(fill=my_cols)
```



Q17. What is the worst ranked chocolate candy?

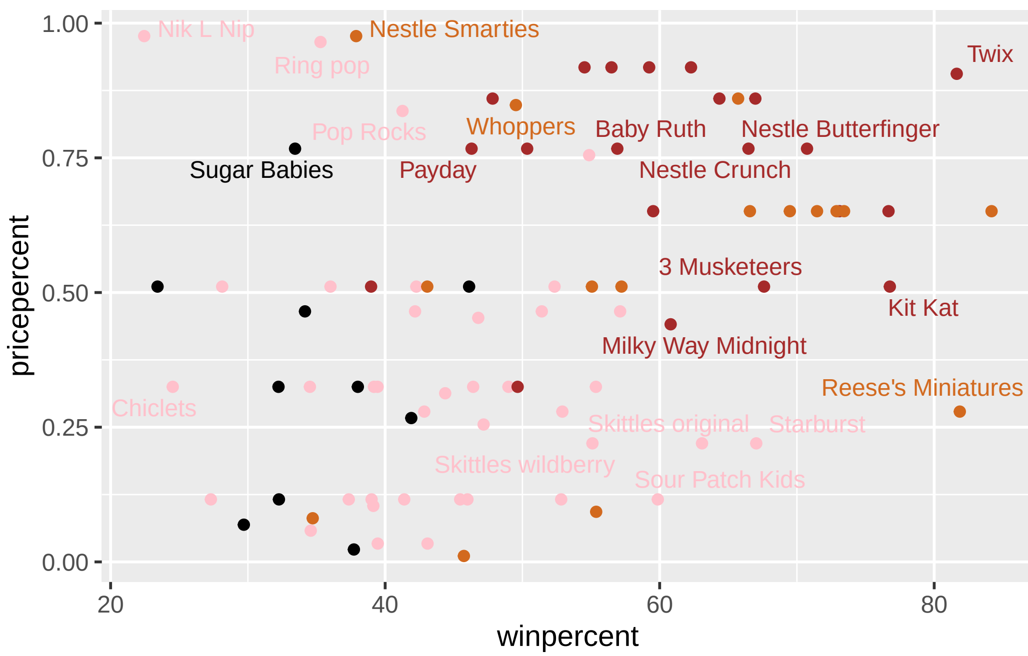
Sixlets

Q18. What is the best ranked fruity candy?

Starburst

```
ggplot(candy.data) +
  aes(winpercent, pricepercent, label=rownames(candy.data)) +
  geom_point(col=my_cols) +
  geom_text_repel(col=my_cols, size=3.3, max.overlaps = 5)
```

```
Warning: ggrepel: 65 unlabeled data points (too many overlaps). Consider
increasing max.overlaps
```



> Q19. Which candy type is the highest ranked in terms of winpercent for the least money - i.e. offers the most bang for your buck?

Chocolate - Reese's Miniatures

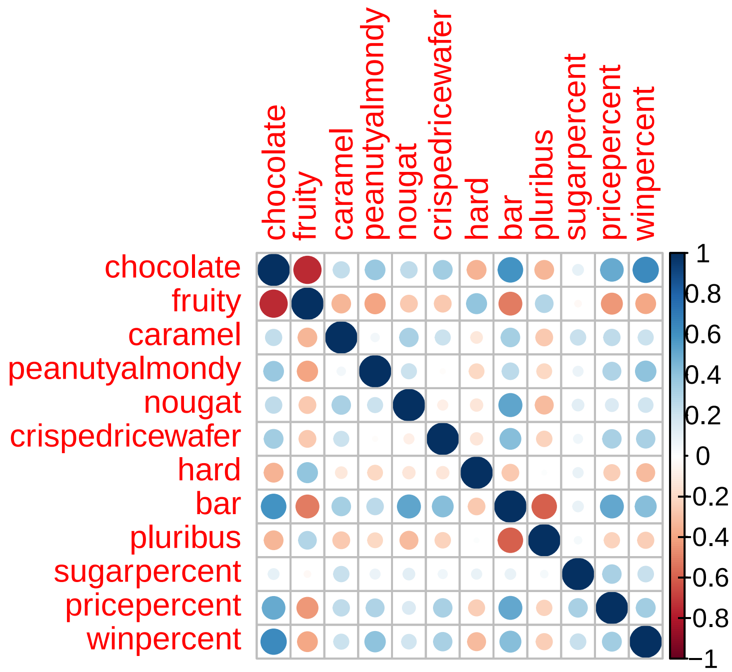
> Q20. What are the top 5 most expensive candy types in the dataset and of these which is the least popular?

The least popular is Nik L Nip

```
candy.data |> arrange(-pricepercent) |> head(5) |> select(winpercent) |> arrange(winpercent)
```

```
                          winpercent
Nik L Nip                   22.44534
Ring pop                    35.29076
Nestle Smarties             37.88719
Hershey's Milk Chocolate    56.49050
Hershey's Krackel           62.28448
```

```
cij <- cor(candy.data)
corrplot(cij)
```



> Q22. Examining this plot what two variables are anti-correlated (i.e. have minus values)?

Fruity and chocolate

> Q23. Similarly, what two variables are most positively correlated?

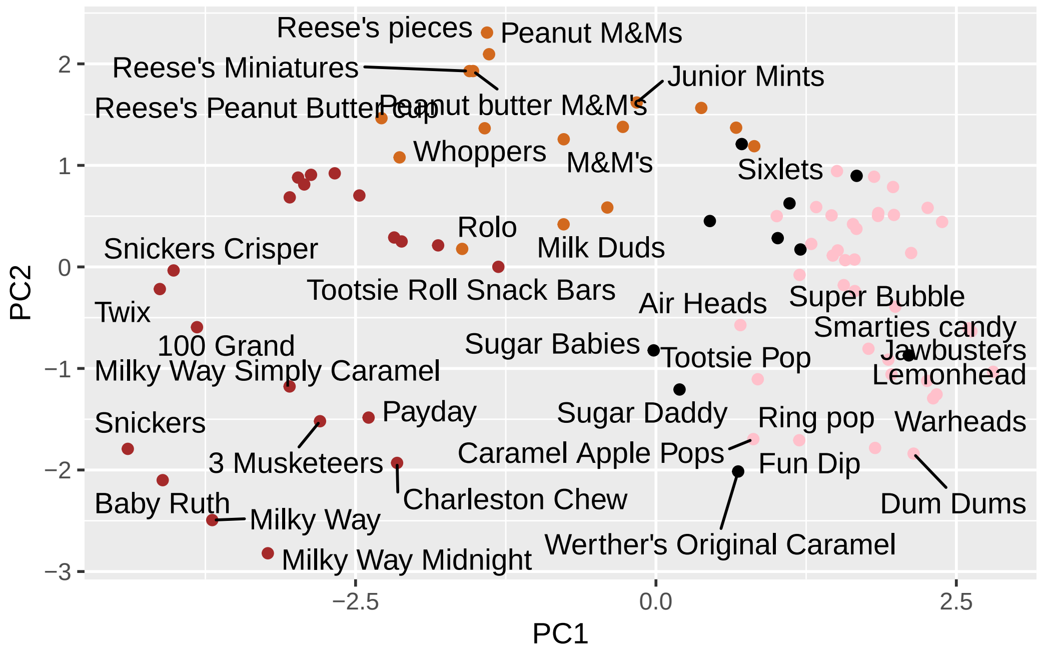
Chocolate and winpercent

```
candy.pca <- prcomp(candy.data, scale=T)
summary(candy.pca)
```

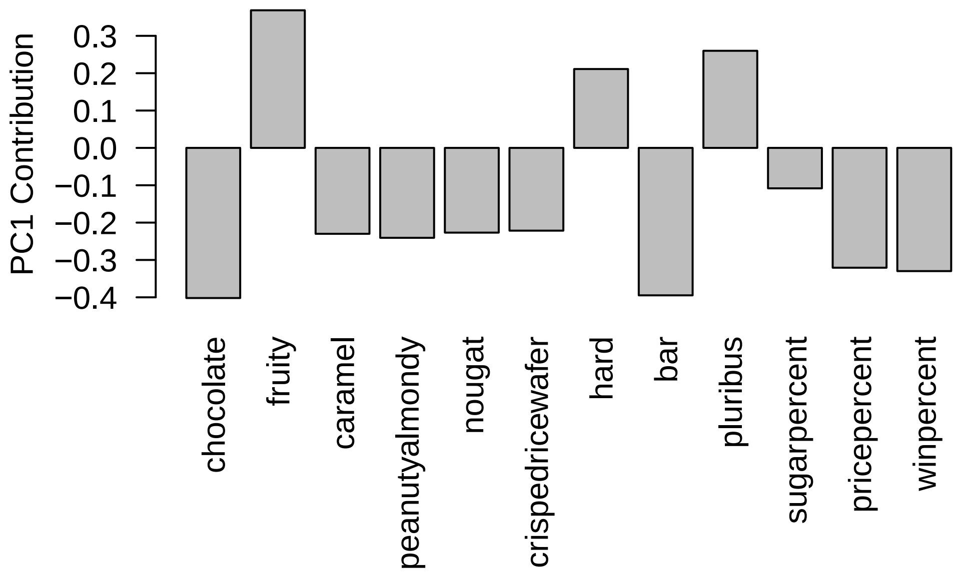
```
Importance of components:
                          PC1    PC2    PC3     PC4    PC5     PC6     PC7
Standard deviation     2.0788 1.1378 1.1092 1.07533 0.9518 0.81923 0.81530
Proportion of Variance 0.3601 0.1079 0.1025 0.09636 0.0755 0.05593 0.05539
Cumulative Proportion  0.3601 0.4680 0.5705 0.66688 0.7424 0.79830 0.85369
                           PC8     PC9    PC10    PC11    PC12
Standard deviation     0.74530 0.67824 0.62349 0.43974 0.39760
Proportion of Variance 0.04629 0.03833 0.03239 0.01611 0.01317
Cumulative Proportion  0.89998 0.93832 0.97071 0.98683 1.00000
```

```r
ggplot(candy.pca$x) +
  aes(PC1, PC2) +
  geom_point(col=my_cols) +
  geom_text_repel(label=rownames(candy.pca$x))
```

```
Warning: ggrepel: 48 unlabeled data points (too many overlaps). Consider
increasing max.overlaps
```




```r
par(mar=c(8,4,2,2))
barplot(candy.pca$rotation[,1], las=2, ylab="PC1 Contribution")
```



Q24. What original variables are picked up strongly by PC1 in the positive direction? Do these make sense to you?

Fruity, hard, pluribus.